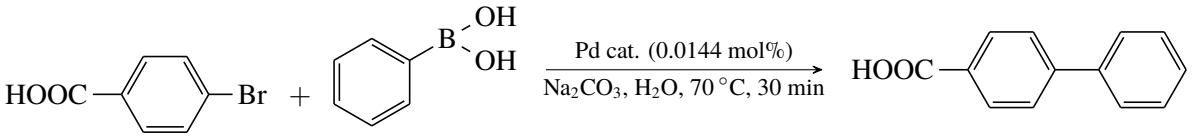
Aqueous Suzuki–Miyaura Cross-Coupling: Synthesis of 4-Phenylbenzoic Acid

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Thursday laboratory, fume hood B7 • 1 October 2026 • Instructor: Dr Hoang Truong Giang

Abstract. 4-Bromobenzoic acid was coupled with phenylboronic acid in aqueous Na_2CO_3 using 0.0144 mol% of a water-soluble Pd–pyrimidine catalyst (70 °C, 30 min). Acidification and recrystallisation from HCl/ethanol gave 4-phenylbenzoic acid as a white, fluffy solid (0.1162 g, 33.6%), identified by ^{1}H NMR and IR spectroscopy. ^{1}H NMR quantification of residual ethanol, acetone and about 0.7 mol% of ethyl 4-phenylbenzoate gave a product content of 98.6 wt% and a corrected yield of 33.2% (TON 2.31×10^3, average TOF 77.1 min^{-1}). The ester and much of the yield loss are attributed to prolonged boiling and excess ethanol during recrystallisation.

1. Introduction

The Suzuki–Miyaura reaction couples an organoboron reagent with an organic halide through oxidative addition of the halide to Pd(0), transmetalation of the organic group from boron to palladium, and reductive elimination of the product (Figure 1).[1] Base is required for transmetalation, either by forming a boronate or by converting Ar–Pd–X into Ar–Pd–OH, which then reacts with the neutral boronic acid; the latter pathway was supported by Carrow and Hartwig for aqueous weak-base conditions.[2] Conventional procedures use organic solvents such as toluene or DMF. Here the reaction is run in water with a phosphine-free catalyst prepared from $Pd(OAc)_2$ and 2-amino-4,6-dihydroxypyrimidine,[3] following an undergraduate procedure that runs in air and isolates the product without extraction.[4,5] The aims were to prepare and purify 4-phenylbenzoic acid, confirm its identity and impurity profile by ^{1}H NMR and IR spectroscopy, and determine the yield and catalyst turnover.



$$Pd^0L_n \xrightarrow{Ar\text{–}Br} Ar{-}Pd^{II}L_n{-}Br \xrightarrow{OH^-} Ar{-}Pd^{II}L_n{-}OH \xrightarrow{PhB(OH)_2} Ar{-}Pd^{II}L_n{-}Ph \xrightarrow{-\,Ar\text{–}Ph} Pd^0L_n$$

Figure 1: Reaction (top) and simplified catalytic cycle via the oxo-palladium pathway (bottom); Ar = 4-carboxylatophenyl. Precatalyst activation is omitted.

2. Procedure

4-Bromobenzoic acid (0.3502 g, 1.742 mmol) and phenylboronic acid (0.2491 g, 2.043 mmol, 1.17 equiv) were stirred with aqueous Na_2CO_3 (0.5 M, 10 mL, 5.0 mmol) in a 20 mL vial until dissolved. Catalyst solution (0.25 mM Pd, 1.0 mL, 0.250 μmol Pd; prepared by staff[5]) was added and the capped vial was heated at 70 °C for 30 min; no Pd black was observed. The cooled mixture was transferred to a 250 mL conical flask in an ice bath and 1 M HCl (20 mL) was added dropwise with stirring (pH paper: approx. 0–1). Water (20 mL) was added, stirring was continued for 10 min, and the crude solid was collected by suction filtration. The solid was recrystallised from 1 M HCl (3 mL) and ethanol (10 mL) with further hot ethanol; the mixture was boiled for longer than intended and more ethanol than the minimum was added (volume not recorded). After cooling (room temperature, 10–15 min; ice bath, 15 min), the crystals were collected by suction filtration, washed with cold ethanol and dried under an IR lamp for 15 min to give 4-phenylbenzoic acid as a white, fluffy solid (0.1162 g, 0.5862 mmol, 33.6%). IR spectra were recorded on a PerkinElmer spectrometer and ^{1}H NMR spectra in $CDCl_3$ at 500 MHz (referenced to $CHCl_3$, δ 7.26).

3. Results

Yield. 4-Bromobenzoic acid is limiting (1.742 vs 2.043 mmol), so the theoretical yield (Table 1) is $1.742 \times 10^{-3} \times 198.22 = 0.3453$ g. The isolated mass, $0.9502 - 0.8340 = 0.1162$ g (bag + product minus bag), corresponds to 33.6%. Residual components in the ^{1}H NMR spectrum were quantified with one resolved signal each (Table 2): each integral I was divided by its proton count and multiplied by the molar mass, giving the relative masses m'. The product fraction is $198.22/200.94 = 98.6$ wt%, so the corrected mass is $0.1162 \times 0.9864 = 0.1146$ g (0.5783 mmol) and the corrected yield 33.2%. With 0.250 μmol Pd (0.0144 mol%), TON $= 0.5783 \times 10^{-3}/(0.250 \times 10^{-6}) = 2.31 \times 10^3$ and the average TOF $= 2313/30$ min $= 77.1$ min^{-1}.

Table 1: Reagents, yield and catalyst metrics. Equivalents are relative to 4-bromobenzoic acid.

Substance	Amount	n/mmol	Equiv. / yield
4-Bromobenzoic acid (201.02 g mol^{-1})	0.3502 g	1.742	1.00
Phenylboronic acid (121.93 g mol^{-1})	0.2491 g	2.043	1.17
Na_2CO_3 (0.5 M)	10 mL	5.0	2.87
Pd catalyst solution (0.25 mM)	1.0 mL	0.000250	0.0144 mol%
4-Phenylbenzoic acid, theoretical (198.22 g mol^{-1})	0.3453 g	1.742	–
Isolated	0.1162 g	0.5862	33.6%
NMR-corrected (98.6 wt%)	0.1146 g	0.5783	33.2%
TON; average TOF over 30 min		2.31×10^3; 77.1 min^{-1}	

Table 2: ^{1}H NMR (500 MHz, $CDCl_3$) of the isolated solid. Integrals are relative to signal a = 2.00; $m' = (I/n_H) \times$MW.

δ/ppm	Mult., J/Hz	n_H	I	Assignment	m'
8.19	d, 8.3	2	2.00	a: product, *ortho* to CO_2H	198.22
7.71	d, 8.3	2	2.04	b: product, *ortho* to Ph	
7.66–7.64	m	2	2.07	c: product, terminal Ph *ortho*	
7.50–7.47	m	2	2.05	d: product, terminal Ph *meta*	
7.43–7.40	m	1	1.00	e: product, terminal Ph *para*	
–	–	1	–	CO_2H not observed (exchange broadened)	
4.41	q, ≈7	2	0.0148	OCH_2 of ethyl 4-phenylbenzoate	1.67
8.11; 1.42	d, 8.5; t	2; 3	0.0149; 0.0204	ArH and CH_3 of the same ester (check)	
3.73	q	2	0.0295	ethanol CH_2[6]	0.68
2.17	s	6	0.0381	acetone[6]	0.37
Sum of m' (product fraction $198.22/200.94 = 98.6$ wt%)					200.94

IR ($\tilde{\nu}$/cm^{-1}):[7] 3300–2500 (br, H-bonded acid O–H), 1674 (s, conjugated C=O), 1607, 1487, 1449 (aromatic C=C), 1316–1286 (C–O), 861 (*para*-disubstituted ring), 747 and 695 (monosubstituted phenyl). Spectra and the datasheet are in the Appendix.

4. Discussion

4.1 Why the reaction works in water

The lab manual asks what makes a reaction in water difficult and why this one works. Most aryl halides and Pd–phosphine catalysts are poorly soluble in water, so a typical coupling would be heterogeneous, and boronic acids can protodeboronate in aqueous base.[8] Here, Na_2CO_3 converts 4-bromobenzoic acid into its water-soluble carboxylate and shifts phenylboronic acid towards the boronate $[PhB(OH)_3]^-$, so both partners dissolve before the catalyst is added. The pyrimidine ligand makes the Pd catalyst water-soluble,[3] and the base also drives transmetalation (Figure 1). The same acid–base handle allows isolation without extraction: HCl destroys carbonate (CO_2 evolution) and protonates the product, which precipitates while boron and inorganic salts stay in water. The method therefore depends on an ionisable substrate; a neutral, insoluble aryl halide would not behave the same way.

4.2 Identity of the product and by-products

The two 2H doublets at δ 8.19 and 7.71 (J = 8.3 Hz) show a *para*-disubstituted ring bearing CO_2H and phenyl, and the 2:2:1 multiplets at δ 7.66–7.40 show the terminal phenyl. The integrals (2.00:2.04:2.07:2.05:1.00) match the 2:2:2:2:1 pattern and the shifts agree with literature values for 4-phenylbenzoic acid in $CDCl_3$.[9] The low-field doublet a distinguishes the product from phenylboronic acid and biphenyl, which also contain a monosubstituted phenyl ring. In the IR spectrum the broad O–H band and the C=O band at 1674 cm^{-1} confirm a conjugated carboxylic acid, and the bands at 861, 747 and 695 cm^{-1} support both ring substitution patterns.

Three weak signals at δ 4.41 (q), 1.42 (t) and 8.11 (d) give nearly equal integrals per proton (0.0074, 0.0068 and 0.0075; Table 2), indicating one compound. They fit ethyl 4-phenylbenzoate: an ethoxy group on an ester oxygen and an aromatic doublet *ortho* to the ester carbonyl; its other aromatic signals overlap with b–d, which explains why those integrals slightly exceed 2.00. The ester (about 0.7 mol%) most likely formed by acid-catalysed Fischer esterification[10] while the product was boiled for longer than intended in HCl-containing ethanol. Ethanol is residual recrystallisation and wash solvent. Acetone is not used in the procedure but is used routinely to rinse glassware before oven drying; incompletely evaporated residue is the likely source, either in the solid or in the NMR sample. These assignments are tentative because they rest on chemical shifts and multiplicities alone. The 98.6 wt% figure is an estimate: the 1.0 s relaxation delay was not optimised for quantitative NMR, and water and inorganic salts are not detected. No unreacted 4-bromobenzoic acid was evident in the aromatic region.

The mechanism also predicts by-products that were not detected directly. Reduction of the Pd(II) precatalyst to Pd(0) is commonly accompanied by homocoupling of two boronic acid molecules, and Pd-catalysed oxidation by air can cause further homocoupling.[8] Biphenyl formed by catalyst activation is limited to about the amount of Pd (0.25 μmol, 0.01% of the product), but air-driven homocoupling is not limited in this way. Biphenyl is neutral and would co-precipitate on acidification, and its signals (δ 7.6–7.3) would overlap with c–e, so a small amount cannot be excluded; it may contribute to the integral sum of 9.16. Protodeboronation converts phenylboronic acid into volatile benzene and reduces the effective excess of the boronic acid; the 1.17 equivalents used provide a margin for this.[8] Hydrodehalogenation would give benzoic acid, which would appear as a separate low-field doublet; apart from the ester doublet at δ 8.11, no other such signal was observed above about 0.4% of the height of signal a.

4.3 Yield and sources of error

The corrected yield (33.2%) is only 0.4 percentage points below the isolated yield, so residual solvent does not explain the low yield. The main method error is in the recrystallisation. Excess hot ethanol leaves more product dissolved in the cold mother liquor, and the cold-ethanol wash dissolves more. The prolonged boiling also converted a small fraction of product into the ester. Transfer losses between the vial, conical flask and filters add to this. Incomplete coupling cannot be excluded either, because the crude solid was not weighed or analysed. The absence of Pd black suggests the catalyst did not visibly decompose, but it does not prove complete conversion. Finally, the 15 min drying did not establish constant mass, so retained water could inflate the mass.

4.4 Improvements, catalyst performance and green aspects

Adding hot ethanol dropwise only until dissolution, recording the total volume, and keeping the boiling period short would reduce both recrystallisation loss and esterification.[10] Weighing the crude solid, or checking it by TLC or ^{1}H NMR before recrystallisation, would separate reaction losses from purification losses. Collecting a second crop from the mother liquor could also recover product. At 0.0144 mol% Pd, the TON of 2.31×10^3 shows a highly efficient catalyst even at this yield. Because TON is normalised to all Pd added and based on isolated product, it is a lower bound on the turnovers achieved in solution.[11] The process uses water as the reaction solvent, runs in air and avoids extraction solvents. Its atom economy, based on the two coupling partners, is $198.22/(201.02+121.93) = 61.4\%$, since HBr and boric acid end up as waste; counting the base would lower it further.[12] In practice, the ethanol used in recrystallisation, the excess base and acid, and the

Pd-containing aqueous filtrate generate far more waste than the reaction stoichiometry, so the solvent saving achieved in the reaction is partly lost during purification.

5. Conclusion

4-Phenylbenzoic acid was prepared by aqueous Suzuki–Miyaura coupling and identified by ^{1}H NMR and IR spectroscopy. The isolated yield was 33.6%, or 33.2% after correction for residual ethanol, acetone and 0.7 mol% of ethyl 4-phenylbenzoate (98.6 wt% product). The objective of making and characterising the product was achieved with good purity but low yield. The low yield and the ester are both consistent with excess ethanol and prolonged boiling during recrystallisation. The catalyst achieved a TON of 2.31×10^3 at 0.0144 mol% Pd, showing that the coupling works efficiently in water when the substrate can be ionised.

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A. NMR spectra and processing record

The figures below were plotted from the Bruker processed spectrum. No additional phasing, baseline correction, smoothing or removal of impurity peaks was applied. Only a uniform chemical-shift correction and intensity scaling were used for display. The original solvent peak at 7.283795 ppm was set to 7.260000 ppm. The 65 536-point processed spectrum has a digital spacing of 0.153 Hz; the reported doublet couplings were calculated from the peak-list separations using 500.13 MHz.

Aromatic integration windows (ppm): a 8.250–8.140; b 7.760–7.675; c 7.675–7.600; d 7.540–7.455; e 7.455–7.350, with a normalised to 2.00. The integral sum is 9.16 rather than a forced 9.00. Shifting all integration windows by ± 0.005 ppm changed any normalised group area by less than 0.01, but this numerical stability does not validate quantitative purity or correct for relaxation effects.

The low-intensity signals in Table 2 (windows 4.45–4.37, 1.45–1.39, 8.13–8.10, 3.76–3.69 and 2.19–2.16 ppm) sit on a broad baseline rise between about δ 1.5 and 2.1 (panel B). Each was therefore integrated above a local straight baseline drawn between the median intensities of 0.02 ppm windows on either side. Changing these flank windows to 0.01 or 0.04 ppm changed the product mass fraction by less than 0.1 wt%.

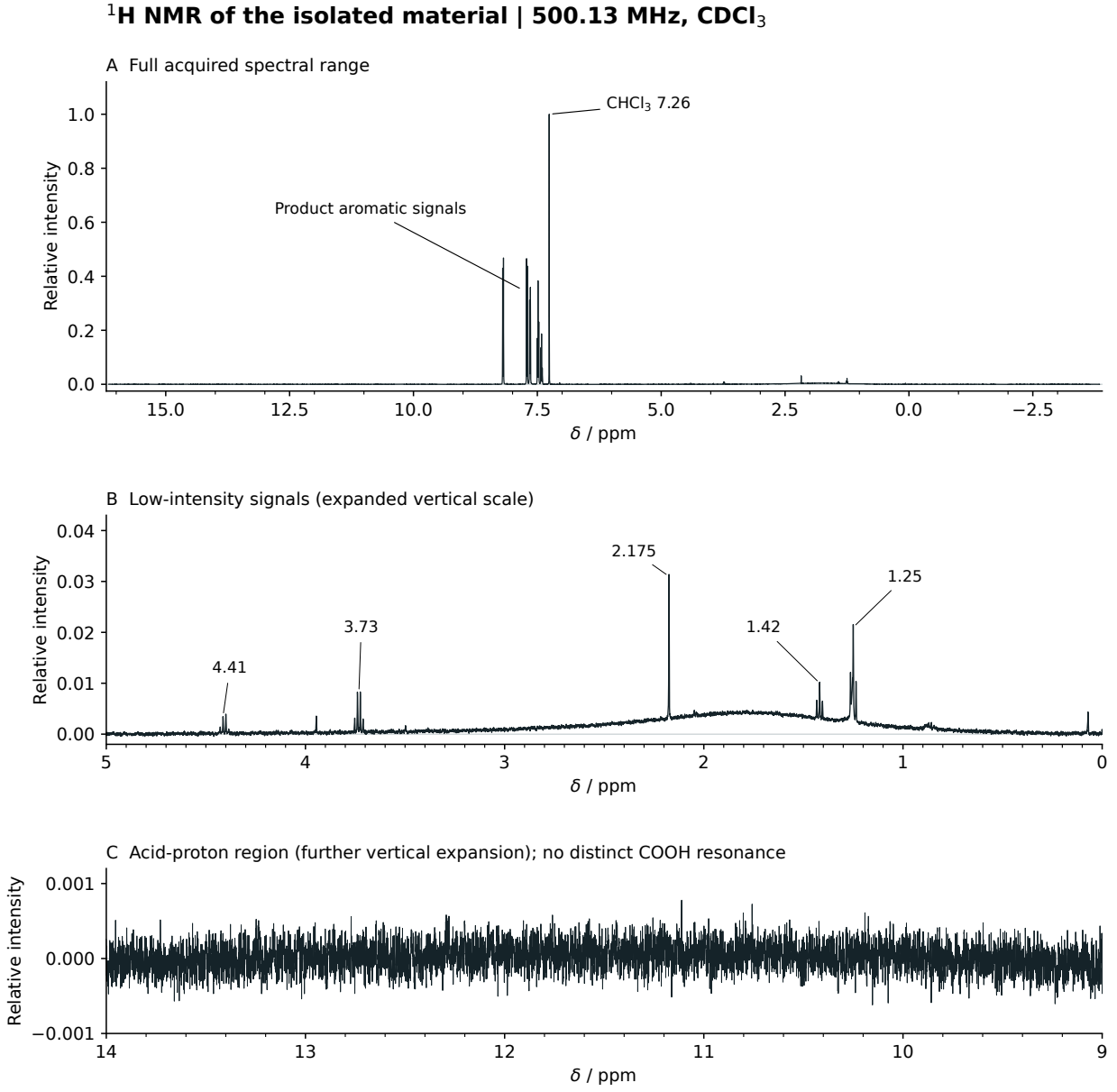


Figure 2: Measured ^{1}H NMR spectrum with expanded views of low-intensity signals and the acid-proton region. Vertical scales differ between panels. Small extra peaks are retained.

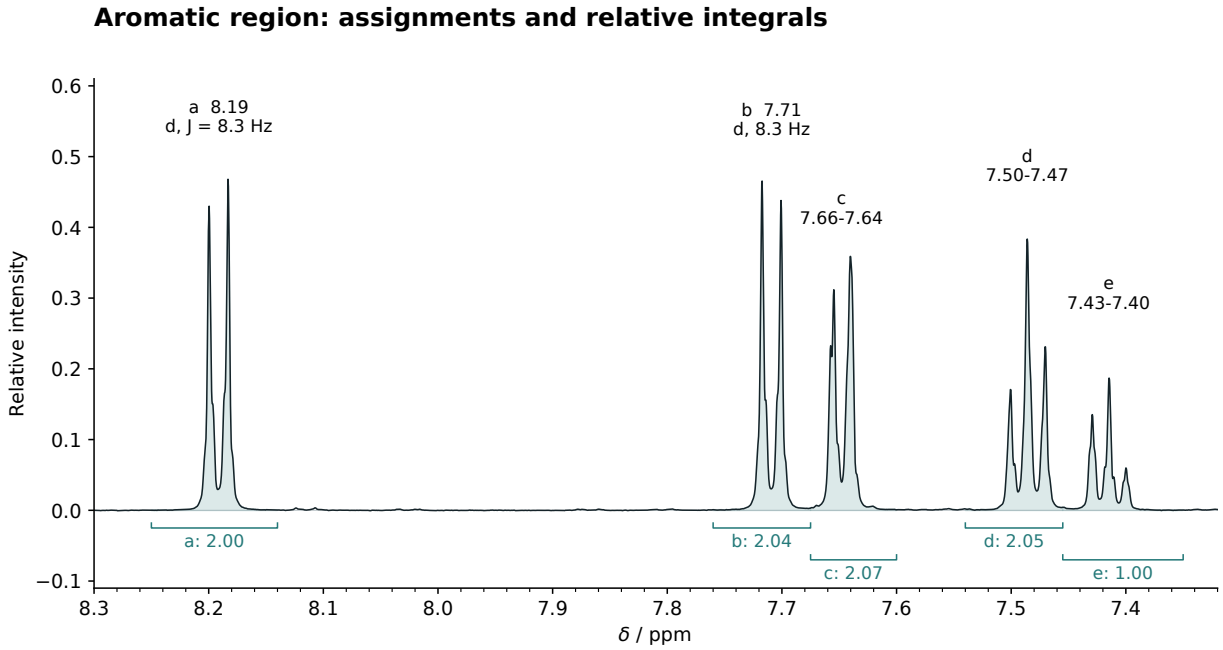


Figure 3: Aromatic-region expansion with assignments a–e, integration windows and measured relative areas. The $CHCl_3$ reference at 7.26 ppm lies outside this enlargement.

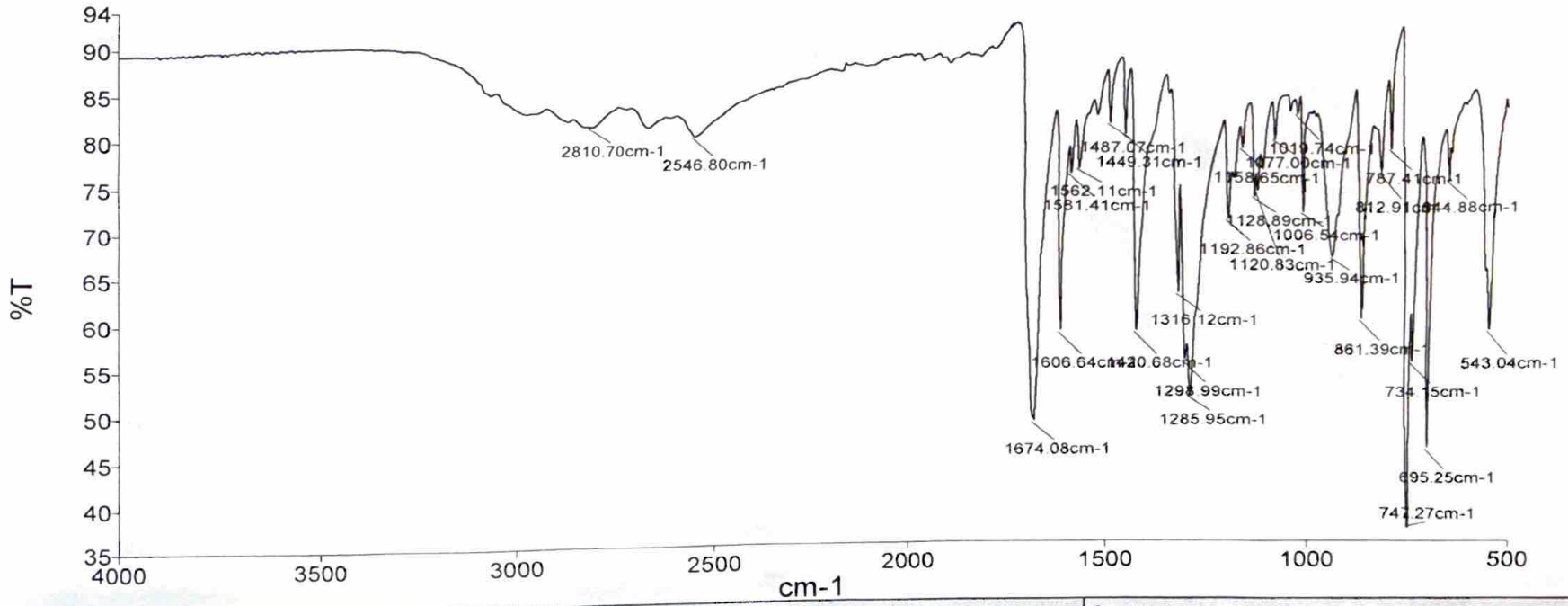
B. Original experimental records

The following pages reproduce the IR printout and the datasheet. The IR page has been rotated for readability; its spectral content has not been altered.

PerkinElmer Spectrum IR Version 10.6.1
Thursday, 1 October, 2026 2:35 PM

Analyst Administrator

Date Thursday, 1 October, 2026 2:35 PM



Sample Name	Description	Quality Checks
Suzuki_KimDongHwan	Sample 980 By Administrator Date Thursday, October 01 2026	The Quality Checks do not report any warnings for the sample.

DATASHEET — WORKING LOG

Suzuki–Miyaura Cross-Coupling in Water

LHN

Name: KIM DONGHWAN Fumehood: B7 Date: 1 Oct 2026 Photos taken: ☐

1. Masses

4-Bromobenzoic acid (g)	0.3502	Phenylboronic acid (g)	0.2491
n(ArBr) = mass/201.02 (mmol)		Equiv. $PhB(OH)_2$	
Empty plastic bag (g)	0.8340	Bag + product (g)	0.9502
Product (g)		Theoretical (g) = $0.9861\ m_{ArBr}$	
		Percentage yield (%)	

2. Time log

Pd added (clock)	Out of block	HCl addition done	Into IR lamp

3. Checkpoints

Stage	**Expected** [source]	**Observed / time**	✓
Solids + Na_2CO_3, stirred	all reactants dissolve [M]; clear, colourless [E]		
Pd catalyst solution	pale yellow [M, App. 1]		
After 30 min at 70 °C	*record colour and any solid or grey tint (Pd black)* [E]	no Pd black solids	
HCl addition	gas forms [M] (CO_2 [C]); solid precipitates [E]		
pH paper after HCl	acidic [M]; calc. pH ≈ 0.7 [C]	0 ~ 1 ?	
Hot ethanol, solution 1	all material dissolves with ~~minimal ethanol~~ [M]	boiled longer ~> eventually more EtOH	
Cooling to RT, then ice	crystals form (recrystallisation) [M]	& reason of small yield?	
Dried product	white solid [E]		

Sources: [M] stated in the lab manual; [C] calculated from manual quantities; [E] general expectation, not stated in the manual. The Expected column is a **checkpoint, not an answer**: record what you actually see; a mismatch is a result worth discussing.

4. Ethanol used in recrystallization & catalyst metrics

Hot EtOH added from sol. 2 (mL, est.)		n(product) (mmol)	
TON = $n_{product}/(2.5 \times 10^{-4}$ mmol)		TOF = TON/30 min (1/min)	
IR file name		NMR tube no. / solvent	

5. Other observations

To be completed at the end of the session		
Student signature	GTA name	GTA signature

This is a working log. The **official** datasheet (lab manual p. 7) must still be printed, signed by the GTA and submitted with the report.